



CoGig Cooperative Platform

Democratic Cooperative Gig Services Platform

Comprehensive Feature Specification, Technical Architecture,
Multi-Objective Algorithmic Models & Workflow Documentation

93%DIRECT WORKER
SHARE**5%**MEMBER WELFARE
FUND**0.12**GINI INCOME
DISPARITY**0%**SURGE PRICE
MARKUP

Cooperative Society: Sri Basaveshwara Labour
Cooperative Society

Jurisdiction Hub: Davangere Urban Hub (15 km
Operating Radius)

Governing Framework: Multi-State Cooperative
Societies Act & ICA Principles

System Version: v2.4 (Production Validated)

Build Artifact: 6,607 Modules · 0 Errors

Target: Hackathon Grand Finale Evaluation

1. Executive Summary & Problem Landscape

1.1 The Gig Economy Crisis

Over 12 million informal service workers in India (plumbers, electricians, carpenters, painters, and domestic help) face systemic economic disenfranchisement on private venture-backed gig platforms (e.g., Urban Company, TaskRabbit). These platforms operate as extractive monopolies characterized by:

- **Extractive Take-Rates (20% to 30%):** Workers lose nearly a third of their gross earnings in platform commissions and lead generation fees.
- **Opaque Algorithmic Deplatforming:** Workers are permanently terminated without human recourse or due process based on unverified customer star ratings.
- **Zero Social Security:** Classified as "independent contractors," gig workers lack health insurance, accidental death cover, or emergency liquidity.
- **Artificial Surge Price Gouging:** Platforms exploit consumer distress during extreme weather and emergencies by hiking prices 1.5x to 3.0x, while capturing the surge windfall rather than passing it to workers.

1.2 The CoGig Cooperative Solution

CoGig replaces extractive corporate middlemen with a **worker-owned platform cooperative** governed by the International Cooperative Alliance (ICA) Principles and legally registered under the Multi-State Cooperative Societies (MSCS) Act. CoGig guarantees:

93% Direct Pay

The vast majority of every rupee paid by customers is settled directly and instantaneously into the worker's verified UPI account upon milestone sign-off.

5% Welfare Fund

Held in a member-governed trust pool providing ₹1,00,000 health insurance, ₹2,00,000 accidental life coverage (PMSBY), and emergency micro-loans.

2% Cost Recovery

Covers bare cloud server costs. All audited year-end surpluses are refunded to active workers as a **Patronage Dividend (सहकारी लाभांश)**.

2. System Architecture & National DPI Integrations

2.1 Core Architectural Layers

CoGig is built on a modern, decoupled micro-architecture combining sub-second React client responsiveness with progressive offline capabilities:

Layer	Technology	Role & Architecture Benefit
Client Layer	React 18 + Vite 8 + Tailwind CSS	Fast single-page application with sub-300ms page transitions, custom semantic tokens, and accessible tap targets.
GIS & Mapping	Leaflet + OpenStreetMap	Zero-cost, privacy-respecting real-time worker pin geolocation, hub serviceability bounding boxes, and routing.
Reactive Engine	AppStateContext + Custom Hooks	Unified state machine orchestrating customer bookings, admin dispatch queues, and worker status in real time.
Storage & Auth	Supabase PostgreSQL + RLS	Encrypted milestone escrow ledger, worker registry, dispute triage tracking, and verifiable proof-of-work storage.
PWA / Offline	Vite PWA Workbox ServiceWorker	Enables offline job inspection, precaches vital assets, and allows standalone home-screen mobile installation.

2.2 National Digital Public Infrastructure (DPI) Integrations

Ministry of Labour & Employment

e-Shram UAN Integration

Every cooperative worker is registered with their 12-digit Universal Account Number (UAN) (e.g. 2847 9102 4821). This connects workers to central social security databases and enables portability across state boundaries.

Ministry of Skill Development

Skill India (NSQF Level 4) PMKVY

National Skills Qualification Framework (NSQF Level 4) accreditation certifies rigorous trade competence. Displayed transparently on booking cards to give customers verified trust.

3. Detailed Stakeholder Portals & Feature Specifications

3.1 Customer Portal (`/customer/\* `)

Key Features Implemented:

- **Davangere Regional Hub Focus:** Configured for Davangere Urban Hub under Sri Basaveshwara Labour Cooperative Society.
- **0% Markup Surge Alert Banner:** Sleek ~80px card displaying pre-monsoon pipe burst alerts with live pulse indicator, zero price markup guarantee, and 1-tap book button. Fully dismissible.
- **Streamlined Post-Requirement Engine:**
 - Work selection dropdown with worker count stepper (1 to 10 workers).
 - Job date and preferred time slot selector (Morning 8–12, Afternoon 12–4, Evening 4–8, Flexible).
 - Address lookup with interactive Leaflet serviceability radar map.
 - Transparent AI Escrow Fare breakdown showing the exact 93/5/2 split.
 - "Hire Worker & Send Booking Request" button that instantly creates the contract and routes to Bookings.
- **Milestone Escrow Release & Review System:** Customer verifies job completion and taps "Approve Day 1 & Release Escrow". Instantly unlocks the Review Worker modal (1–5 stars, compliment chips, and testimonial textarea).
- **Cash Demand Whistleblower Reporting:** Direct reporting tool if a worker asks for off-platform cash, instantly flagging the ticket to cooperative administrators.

3.2 Gig Worker Portal (`/worker/\* `)

Key Features Implemented:

- **Mobile-First Low-Literacy UI:** 48px minimum touch targets, high contrast typography, and multi-lingual voice accessibility in English, Hindi, and Kannada.
- **Live Availability Toggle:** Real-time toggle publishing GPS availability to the cooperative dispatch engine.
- **Complete Payout Transparency:** Standard daily wage breakdown (₹500 gross: ₹465 direct to worker, ₹25 to welfare, ₹10 to platform).
- **Digital Proof of Work (PoW):** Camera-captured Before/After photos synced to the customer contract for instant dispute prevention.
- **Year-End Patronage Dividend (सहकारी लाभांश) Card:** Implements ICA Principle 3, tracking accumulated dividend (₹4,260 across 142 jobs) to be disbursed at the Annual General Meeting (AGM).
- **Welfare Insurance Coverage:** Shows active ₹1,00,000 hospitalization and ₹2,00,000 PMSBY accidental cover.

3.3 Cooperative Admin Portal (`/coop-admin/\* `)

The Cooperative Administration portal serves as the command center for cooperative office bearers, supervisors, and labor union representatives. It replaces arbitrary automated management with transparent, human-auditable cooperative tools.

Unified Command Center Architecture:

- **Sidebar Navigation:** Dedicated order: [Dashboard](#) → [Workers](#) → [Bookings](#) → [Fair Allocation](#) → [Demand Forecast](#) → [Complaints](#) .
- **Header Bar:** Displays Davangere Hub, MSCS/CR/2026/KA-08 registration, NCDC recognition, language switcher, and admin profile.
- **Executive KPI Dashboard:** 4 core metrics (Rating 4.8★, Welfare Fund ₹5,800, Active Members, Bookings 1,248).

Spatial GIS Telemetry

Live Worker GPS Map

Powered by Leaflet and OpenStreetMap, the live map plots verified worker locations across Davangere Urban Hub (15 km radius), displaying live availability status, trade color badges, and active dispatch markers.

Ethical AI Dispatch

Fair Allocation Engine

Maintains an audited Gini income disparity score of 0.12. Features a sub-30 minute idle time watchdog that elevates waiting members to priority dispatch, eliminating algorithmic starvation.

Predictive AI Analytics

Demand Forecast & Surge Publisher

7-day predictive curves model weather-induced demand surges (e.g. pre-monsoon plumbing leaks). Enables 1-click publishing of zero-markup alerts to mobilize standby crews without price gouging.

Worker Protection

Anti-Deplatforming Guarantee

No worker can be banned by an automated rating algorithm. Disputes follow a 3-tier escalation process: AI Mediation → Conciliation → Elected Cooperative Supervisory Council hearing.

4. End-to-End Workflow & User Journeys

Workflow 1: Customer Booking to Instant Settlement & Review

- 1

Service & Schedule Selection: Customer selects trade (Plumbing), quantity (1 worker), date, and preferred slot (Morning 8–12 AM).
- 2

Location & Fare Estimation: Customer enters address; Leaflet verifies serviceability in Davangere Hub. AI calculates fair wage (₹550/day) with 0% surge markup.
- 3

Escrow Lock & Booking Creation: Customer clicks "Hire Worker & Send Booking Request". Funds are placed into digital escrow; booking appears on customer and admin ledgers.
- 4

Fair Allocation Dispatch: FWA algorithm assigns the most equitable, certified worker (Manjunath Gowda, NSQF-4 certified). Worker accepts via Worker Portal.
- 5

Service Execution & Proof of Work: Worker completes repair, uploads verification photos.
- 6

Escrow Release & Instant UPI Settlement: Customer taps "Approve Day 1 & Release Escrow". 93% (₹511.50) is instantly transferred to worker's UPI, 5% (₹27.50) to Welfare Fund, and 2% (₹11.00) to Platform.
- 7

Worker Review Modal: Modal opens automatically; customer rates 5 stars, selects compliment tags, and submits verified feedback.

Workflow 2: Ethical AI Fair Work Allocation (FWA) Engine

$$\text{Fairness_Score} = (0.30 \times \text{Wait_Time}) + (0.25 \times \text{Income_Parity_Delta}) + (0.20 \times \text{Skill_Match}) + (0.15 \times \text{Distance_Inv}) + (0.10 \times \text{Job_Count_Inv})$$

Component	Weight	Algorithmic Purpose & Social Equity Impact
Wait Time Score	30%	Prioritizes members who have been waiting the longest. Watchdog triggers an emergency +0.40 priority boost if idle > 30 minutes.
Income Parity Delta	25%	Calculates the gap between worker's monthly earnings and the cooperative target living wage, preventing algorithmic superstar concentration.
Skill & NSQF Match	20%	Validates Skill India NSQF-4 certification for specialized trades (e.g. electrical hazards) to ensure citizen safety.
Proximity (Distance Inv)	15%	Minimizes worker commute time and fuel expenses to sustain sub-30 minute SLAs.
Job Count Inverse	10%	Balances the total number of monthly dispatches across the entire cooperative roster.

5. Comparative Evaluation & Hackathon Impact Matrix

5.1 Comprehensive Benchmark: CoGig vs. Industry Alternatives

Evaluation Dimension	CoGig Cooperative Platform	Private Platforms (Urban Company)	Unorganized Contractors
Commission / Take Rate	2% (Cost-recovery; surplus refunded at AGM)	20% – 30% permanent venture extraction	15% – 35% informal middlemen cut
Worker Direct Share	93% guaranteed	70% – 80%	65% – 85% (inconsistent)
Social Security Safety Net	5% Dedicated Welfare Fund (Health, Insurance, Loans)	None (Classed as independent contractors)	Zero safety net
Surge Pricing Policy	0% Markup Guarantee (Standby mobilization)	1.5x – 3.0x Artificial surge markups	Unregulated price gouging
Dispatch Algorithm	Transparent FWA Engine (Gini = 0.12)	Black-box profit-maximizing AI	Subjective personal favoritism
Deplatforming Safeguards	Democratic Guarantee (Elected Council hearing)	Arbitrary algorithmic account blocking	Immediate informal firing
National DPI Integration	e-Shram, Skill India, PMSBY, Aadhaar, UPI	Proprietary closed database	Cash-only, unverified
Ownership & Governance	Worker-Owned Cooperative (1 Member = 1 Vote)	Corporate Shareholders / VC Equity	Unregistered proprietary entity

5.2 Economic Impact on Worker Livelihoods

Worker Net Earnings Increase

By reducing platform fees from 25% to 2%, a worker completing 25 gigs a month at ₹550/day increases their monthly take-home income from ₹10,312 to ₹12,787 — an immediate **+24.0% increase in net disposable wages**.

Patronage Dividend & Welfare Wealth

In addition to direct earnings, the worker accumulates **₹687/month in democratic welfare reserves** and an estimated **₹4,260 in annual Patronage Dividends (सहकारी लाभांश)** disbursed at the AGM under ICA Principle 3.

5.3 Hackathon Feasibility & Future Scaling Roadmap

- **Phase 1 (Completed):** Full-stack cooperative web portal (Customer, Worker, Admin), 93/5/2 escrow engine, Leaflet GIS mapping, FWA algorithmic fairness, and DPI badge verification.
- **Phase 2 (Next 60 Days):** WhatsApp Vernacular Conversational Bot integration via Twilio for voice-based dispatch in local Kannada and Hindi dialects.
- **Phase 3 (Next 180 Days):** Integration with Open Network for Digital Commerce (ONDC) to allow any buyer app (Paytm, PhonePe) to discover CoGig cooperative crews.